

Andreas Eglhofer

eglhofe.andreas@gmail.com | +1 (864) 907-4270 | linkedin.com/in/andreaseglhofer | GitHub | Personal Website

Education

Massachusetts Institute of Technology

Cambridge, MA

B.S. in Physics, Expected 2030

South Carolina Governor's School for Science and Mathematics

Hartsville, SC

High School Diploma, 2026

GPA: 4.00/4.00 unweighted, 5.566/5.00 weighted

SAT: 1550, 800 Math, 750 EBRW

Relevant Coursework: Single/Multivariable Calculus, Mechanics, Electricity and Magnetism, Discrete Structures, Number Theory, Modern Physics, Fluids, Thermodynamics, Optics, Data Structures and Algorithms

Research Experience

Research Assistant, Max Planck Institute for Software Systems

Kaiserslautern, Germany

June 2025 – August 2025

- Researched FunSearch-style methods using large language models for combinatorial optimization and search problems.
- Implemented and evaluated approaches on tasks including MaxSAT and word separation.
- Presented technical work in an international research setting under Dr. Rupak Majumdar

Individual Projects

Ultimate Tic Tac Toe AI

Python

- Built an AI for Ultimate Tic Tac Toe using depth-limited minimax with alpha-beta pruning, iterative deepening, and move ordering.
- Designed a heuristic evaluation function combining mini-board control, macro-board line scoring, center and corner weighting, and active-board pressure.
- Implemented state normalization, legal-move generation, terminal-state detection, and time-capped search to improve decision quality under runtime constraints.

Guard Shift

andreas-eg.itch.io/guard-shift

- Built a top-down stealth game in Python and Pygame with guard behavior driven by A* search on a grid-based state space.
- Implemented dynamic path replanning across patrol, investigate, chase, and search states using Manhattan-distance heuristics and walkable-tile successor generation.
- Designed interactive mechanics including noise-triggered guard responses, door-based path blocking, alert escalation, and multi-level gameplay.

Pressurized Water Reactor Dynamics Simulator

pwr-simulator.vercel.app

- Built an interactive simulator modeling reactor power output, core temperature, control rod motion, thermal feedback, and SCRAM logic.
- Researched reactor physics independently and translated physical system behavior into a computational model.
- Developed the application in JavaScript and React.

Technical Skills

Languages: Python, C++, Bash, LaTeX

Libraries: Pygame, React, NumPy, SciPy, pandas, Matplotlib

Tools: Git, Linux, Neovim, Nix, CMake, SSH, KiCad, SolidWorks

ML: PyTorch, TensorFlow

Honors

President's Cup (2026) – GSSM's only graduation award, given in place of class rank to the senior who best embodies the school's highest ideals and aspirations.

Physics Departmental Award (2026) – Recognized by the Physics Department for exceptional performance and engagement in advanced physics coursework.

George Eastman Young Leaders Award (2025) – Selected for academic excellence, leadership, and meaningful contributions to school and community.